

SIMATS ENGINEERING ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour

Total Marks: 30

Annexure A Resolution Algorithm Implementation

Code of Conduct:

I J O Joanna Divine (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Reg No: 192572003

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: JOANNA

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Let: **P** = It rains, **Q** = The ground is wet.

1. Propositional Logic Representation

Premise 1: $P \rightarrow Q$ (If it rains, the ground becomes wet)

Premise 2: **P** (It is raining)

Goal: **Q** (The ground is wet)

2. Conversion to CNF

$P \rightarrow Q \equiv \neg P \vee Q$

Clause 1: $\neg P \vee Q$

Clause 2: **P**

3. Negation of Goal

$\neg Q$ is added to the clause set as Clause 3.

4. Resolution Steps

Step 1: Resolve Clause 2 (**P**) with Clause 1 ($\neg P \vee Q$) $\rightarrow$ resolvent: **Q**

Step 2: Resolve **Q** (Step 1) with Clause 3 ($\neg Q$) $\rightarrow$ resolvent: **■** (Empty Clause)

5. Conclusion

Since the Empty Clause (**■**) is derived, the negated goal leads to a contradiction. Hence, the goal "**the ground is wet (Q)**" is **proved true** by the Resolution Algorithm.

Question 2 – Student Assignment Submission

Let: **S** = A student submits the assignment, **M** = The student receives internal marks.

1. Propositional Logic Expressions

Premise 1: $S \rightarrow M$

Premise 2: **S(Rahul)** — Rahul submitted the assignment

Goal: **M(Rahul)** — Rahul receives internal marks

2. Conversion to CNF

$S \rightarrow M \equiv \neg S \vee M$

Clause 1: $\neg S \vee M$

Clause 2: **S**

3. Negation of Goal

$\neg M$ is added as Clause 3.

4. Resolution Steps

Step 1: Resolve Clause 2 (**S**) with Clause 1 ($\neg S \vee M$) $\rightarrow$ resolvent: **M**

Step 2: Resolve **M** (Step 1) with Clause 3 ($\neg M$) $\rightarrow$ resolvent: **■** (Empty Clause)

5. Conclusion

The Empty Clause (**■**) is obtained, proving a contradiction with the negated goal. Hence, it is **proved** that **Rahul receives internal marks (M)**.

Question 3 – Library Membership

Let: **L** = A person is a library member, **B** = The person can borrow books.

1. Propositional Logic Representation

Premise 1: $L \rightarrow B$

Premise 2: $L(\text{Priya})$ — Priya is a library member

Goal: $B(\text{Priya})$ — Priya can borrow books

2. Conversion to CNF

$L \rightarrow B \equiv \neg L \vee B$

Clause 1: $\neg L \vee B$

Clause 2: L

3. Negation of Goal

$\neg B$ is added as Clause 3.

4. Resolution (performed until no new clauses can be generated)

Step 1: Resolve Clause 2 (L) with Clause 1 ($\neg L \vee B$) $\rightarrow$ resolvent: B

Step 2: Resolve B (Step 1) with Clause 3 ($\neg B$) $\rightarrow$ resolvent: $\blacksquare$ (Empty Clause)

No further clauses can be generated after the empty clause is reached.

5. Final Conclusion

Since the Empty Clause ($\blacksquare$) is derived, the goal is **proved true: Priya can borrow books (B)**.

Question 4 – Placement Eligibility

Let: **A** = A student clears the aptitude test, **E** = The student is eligible for placement.

1. Propositional Logic

Premise 1: $A \rightarrow E$

Premise 2: $A(\text{Arun})$ — Arun cleared the aptitude test

Goal: $E(\text{Arun})$ — Arun is eligible for placement

2. CNF Clauses

$A \rightarrow E \equiv \neg A \vee E$

Clause 1: $\neg A \vee E$

Clause 2: A

3. Negation of Goal

$\neg E$ is added as Clause 3.

4. Resolution Steps

Step 1: Resolve Clause 2 (A) with Clause 1 ($\neg A \vee E$) $\rightarrow$ resolvent: E

Step 2: Resolve E (Step 1) with Clause 3 ($\neg E$) $\rightarrow$ resolvent: $\blacksquare$ (Empty Clause)

5. Resolution Tree

Clause 2 (A) + Clause 1 ($\neg A \vee E$) $\rightarrow$ E

E + Clause 3 ($\neg E$) $\rightarrow$ $\blacksquare$ (Empty Clause)

Conclusion: The derivation of the Empty Clause ($\blacksquare$) proves that the negated goal is unsatisfiable. Hence it is **proved** that **Arun is eligible for placement (E)**.

Question 5 – Access Control System

Let: **C** = The user enters the correct password, **Au** = The user is authenticated, **Ac** = The user is granted access.

1. Propositional Logic Representation

Premise 1: $C \rightarrow Au$

Premise 2: $Au \rightarrow Ac$

Premise 3: C — the user entered the correct password

Goal: Ac — the user is granted access

2. Conversion of Implications to CNF

$C \rightarrow Au \equiv \neg C \vee Au$

$Au \rightarrow Ac \equiv \neg Au \vee Ac$

Clause 1: $\neg C \vee Au$

Clause 2: $\neg Au \vee Ac$

Clause 3: C

3. Negation of Goal

$\neg Ac$ is added as Clause 4.

4. Resolution Steps to Derive the Empty Clause

Step 1: Resolve Clause 3 (C) with Clause 1 ($\neg C \vee Au$) $\rightarrow$ resolvent: Au

Step 2: Resolve Au (Step 1) with Clause 2 ($\neg Au \vee Ac$) $\rightarrow$ resolvent: Ac

Step 3: Resolve Ac (Step 2) with Clause 4 ($\neg Ac$) $\rightarrow$ resolvent: $\blacksquare$ (Empty Clause)

5. Explanation and Final Conclusion

Step 1 combines the fact that the correct password was entered with the rule "password $\rightarrow$ authenticated" to establish that the user is authenticated (Au).

Step 2 combines this authentication result with the rule "authenticated $\rightarrow$ access" to establish that the user is granted access (Ac).

Step 3 resolves this derived fact (Ac) with the negated goal ($\neg Ac$), producing the Empty Clause ($\blacksquare$), which is a contradiction.

Since assuming the negation of the goal leads to a contradiction, the original goal must be true. Hence, it is **proved that the user is granted access (Ac)**.